

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
=====
CONFIGURACIÓN INICIAL
=====
Configuración Valor
      Device  cuda
      Seed    42
Output Width  100
=====
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',           # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 500,                   # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para cl
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments,
    'extras': {
        'expansion': 32,
        'sparsity_coef': 5e-4,
        'epochs': 200,
        'patience': 15,
        'learning_rate': 1e-3,
        'sparsity_decay': 0.98,
```

```

        'batch_size': 512
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')

```

```

Metadata de naming:
variante: standard
tecnica: l1-decay
capa: 6
juegos: 500
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 32, 'sparsity_coef': 0.0005, 'epochs': 200, 'patience': 15, 'learning

```

```

extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')

```

Nuevo extras registrado: ex4 → {'expansion': 32, 'sparsity_coef': 0.0005, 'epochs': 200, 'p
 extras_id: ex4

3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```

# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 16384,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 512,
    'num_epochs': 200,
    'learning_rate': 1e-3,
    'sparsity_coef': 5e-4,
    'sparsity_decay': 0.98,

    # Early Stopping
    'early_stopping': True,
    'patience': 15,

    # Datos
    'activations_file': Path(f"../../activations/data/layer{NAMING_META['capa']}_{NAMING_I"},
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}

# Crear directorio local del modelo (plots_dir ya fue creado por get_plots_dir)
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración con formato profesional
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

```

```

# Arquitectura
for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Entrenamiento
for key in ['batch_size', 'num_epochs', 'learning_rate', 'sparsity_coef', 'sparsity_decay']:
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Early Stopping
for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)

```

```

=====
CONFIGURACIÓN DEL SAE
=====

```

Categoría	Parámetro	Valor
Arquitectura	input_dim	512
Arquitectura	hidden_dim	16384
Arquitectura	tied_weights	False
Entrenamiento	batch_size	512
Entrenamiento	num_epochs	200
Entrenamiento	learning_rate	0.001

```

Entrenamiento sparsity_coef 0.0005
Entrenamiento sparsity_decay 0.98
Early Stopping early_stopping True
Early Stopping patience 15
=====

Archivo de activaciones: ..\..\..\activations\data\layer6_500games.npy
Expansión: 32.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\500games\ex4\plots
=====

```

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})

print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')

```

```

print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

```

```

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False, # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)

```

```

=====
DATASET DE ACTIVACIONES
=====

```

	Propiedad	Valor
	Total muestras	29,500
	Shape (29500, 512)	
	Dtype	float32
Tamaño en memoria		57.62 MB

```

=====

=====
SPLIT DE DATOS (80-20)
=====

```

	Conjunto	Muestras	Porcentaje
	Train	23,600	80.0%
Validation		5,900	20.0%
	Total	29,500	100.0%

```

=====

```

=====				
DATALOADERS				
=====				
Loader	Batches	Batch Size	Shuffle	
Train	47	512	Sí	
Validation	12	512	No	
=====				

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder:** Linear $\rightarrow$ ReLU (sparsity natural)- **Decoder:** Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)

# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim'] # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '_'
    ],
    'Parámetros': [
```

```

        '-',
        f"{encoder_params: ,}",
        '-',
        f"{decoder_params: ,}",
        '-',
        f"{num_params: ,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)

```

```

=====
ARQUITECTURA DEL MODELO
=====

```

	Componente	Dimensión	Parámetros
	Input	512	-
	Encoder	512 → 16384	8,404,992
Hidden	(Latent)	16384	-
	Decoder	16384 → 512	8,389,120
	Output	512	-
	Total	-	16,794,112

```

=====

Factor de expansión: 32.0x
Tied weights: No
Device: cuda
=====

```

6. Función de Pérdida

Loss = MSE(reconstrucción) + λ × L1(activaciones)

- **MSE:** Mide qué tan bien reconstruimos las activaciones originales
- **L1:** Penaliza activaciones grandes → fuerza sparsity

```
def loss_function(x, reconstruction, hidden, sparsity_coef):
    """
    Pérdida compuesta: MSE + L1 penalty

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse)
        sparsity_coef: Peso de la penalización L1
    """
    # Error de reconstrucción
    mse_loss = F.mse_loss(reconstruction, x)

    # Sumar sobre las features (dim=1), promediar sobre el batch
    sparsity_loss = hidden.abs().sum(dim=1).mean()

    # Pérdida total
    total_loss = mse_loss + sparsity_coef * sparsity_loss

    return total_loss, mse_loss, sparsity_loss

# Optimizador
optimizer = torch.optim.Adam(model.parameters(), lr=CONFIG['learning_rate'])

# Resumen del optimizador
optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'Sparsity Decay'],
    'Valor': [
        'Adam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
        f"{CONFIG['sparsity_decay']:.2f}"
    ]
})

print('\n' + '='*50)
print(' OPTIMIZACIÓN')
print('='*50)
print(optim_info.to_string(index=False))
print('='*50)
```

```
=====
OPTIMIZACIÓN
=====
                Parámetro Valor
            Optimizador  Adam
            Learning Rate 1e-03
Sparsity Coef (inicial) 5e-04
            Sparsity Decay 0.98
=====
```

7. Loop de Entrenamiento

```
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
    'sparsity_coef_history': [] # Guardar evolución del coeficiente L1
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' L1 decay: {CONFIG["sparsity_coef"]:.2e} × {CONFIG["sparsity_decay"]:.2f}^época')
print(f' Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):
```

```

# ===== L1 DECAY =====
# Reducir sparsity_coef exponencialmente
current_sparsity_coef = CONFIG['sparsity_coef'] * (CONFIG['sparsity_decay'] ** epoch)
history['sparsity_coef_history'].append(current_sparsity_coef)

# ===== ENTRENAMIENTO =====
model.train()

epoch_total_loss = 0
epoch_mse_loss = 0
epoch_sparsity_loss = 0
epoch_l0 = 0
epoch_fraction_active = 0

# Barra de progreso
pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] L1={current_sparsity_coef}')

for batch in pbar:
    # Mover al device
    x = batch.to(device)

    # Forward pass
    reconstruction, hidden = model(x)

    # Calcular pérdida con sparsity_coef actualizado
    total_loss, mse_loss, sparsity_loss = loss_function(
        x, reconstruction, hidden, current_sparsity_coef
    )

    # Backward pass
    optimizer.zero_grad()
    total_loss.backward()

    optimizer.step()

    # Métricas
    with torch.no_grad():
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

    # Acumular
    epoch_total_loss += total_loss.item()

```

```

epoch_mse_loss += mse_loss.item()
epoch_sparsity_loss += sparsity_loss.item()
epoch_l0 += l0
epoch_fraction_active += fraction_active

# Actualizar barra
pbar.set_postfix({
    'loss': f'{total_loss.item():.4f}',
    'mse': f'{mse_loss.item():.4f}',
    'l0': f'{l0:.1f}'
})

# Promedios de entrenamiento
num_train_batches = len(train_loader)
avg_train_total_loss = epoch_total_loss / num_train_batches
avg_train_mse_loss = epoch_mse_loss / num_train_batches
avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
avg_train_l0 = epoch_l0 / num_train_batches
avg_train_fraction_active = epoch_fraction_active / num_train_batches

# ===== VALIDACIÓN =====
model.eval()

val_total_loss = 0
val_mse_loss = 0
val_sparsity_loss = 0
val_l0 = 0
val_fraction_active = 0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)

        # Forward pass
        reconstruction, hidden = model(x)

        # Calcular pérdida con sparsity_coef actualizado
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, current_sparsity_coef
        )

# Métricas

```

```

10 = (hidden > 0).float().sum(dim=1).mean().item()
fraction_active = (hidden > 0).float().mean().item()

# Acumular
val_total_loss += total_loss.item()
val_mse_loss += mse_loss.item()
val_sparsity_loss += sparsity_loss.item()
val_l0 += l0
val_fraction_active += fraction_active

# Promedios de validación
num_val_batches = len(val_loader)
avg_val_total_loss = val_total_loss / num_val_batches
avg_val_mse_loss = val_mse_loss / num_val_batches
avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
avg_val_l0 = val_l0 / num_val_batches
avg_val_fraction_active = val_fraction_active / num_val_batches

# Guardar en historial
history['train_total_loss'].append(avg_train_total_loss)
history['train_mse_loss'].append(avg_train_mse_loss)
history['train_sparsity_loss'].append(avg_train_sparsity_loss)
history['train_l0_sparsity'].append(avg_train_l0)
history['train_fraction_active'].append(avg_train_fraction_active)

history['val_total_loss'].append(avg_val_total_loss)
history['val_mse_loss'].append(avg_val_mse_loss)
history['val_sparsity_loss'].append(avg_val_sparsity_loss)
history['val_l0_sparsity'].append(avg_val_l0)
history['val_fraction_active'].append(avg_val_fraction_active)

# ===== EARLY STOPPING =====
if CONFIG['early_stopping']:
    if avg_val_mse_loss < best_val_mse:
        best_val_mse = avg_val_mse_loss
        best_epoch = epoch + 1
        epochs_no_improve = 0

    # Guardar mejor modelo (local + organizado)
    best_checkpoint = {
        'model_state_dict': model.state_dict(),
        'optimizer_state_dict': optimizer.state_dict(),

```

```

        'config': CONFIG,
        'history': history,
        'epoch': best_epoch,
        'val_mse': best_val_mse
    }

    # Guardado local
    best_model_name = get_model_name(
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best'
    )
    torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

    # Guardado organizado
    save_checkpoint(
        model,
        base_path=NAMING_META['base_path'],
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=CONFIG,
        history=history,
        optimizer=optimizer
    )

    print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= CONFIG['patience']:
        print(f'\n Early stopping activado en época {epoch+1}')
        print(f' Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}')
        break

# Log cada 5 épocas
if (epoch + 1) % 5 == 0 or epoch == 0:
    print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | L1 coef: {current_sparsity_coef:}')

```

```

        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}')
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0')
        if CONFIG['early_stopping']:
            print(f'    Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {best_val_mse:.4f}')

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

# Tabla de métricas finales
final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})

print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)

```

Iniciando entrenamiento: 200 épocas máximo
 L1 decay: $5.00e-04 \times 0.98^{\text{época}}$
 Early stopping: patience=15 épocas

=====

Época 1/200 [Train] L1=5.00e-04: 100%| | 47/47 [00:01<00:00, 40.90it/s, loss=0.4550, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 1 (Val MSE: 0.3168)

Época 1/200 | L1 coef: 5.00e-04

Train - Total: 0.6495 | MSE: 0.5376 | L0: 1289.4

Val - Total: 0.4289 | MSE: 0.3168 | L0: 936.6

Best Val MSE: 0.3168 @ época 1 | Sin mejora: 0/15

Época 2/200 [Train] L1=4.90e-04: 100%| | 47/47 [00:00<00:00, 66.60it/s, loss=0.2611, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 2 (Val MSE: 0.1629)

Época 3/200 [Train] L1=4.80e-04: 100%| | 47/47 [00:00<00:00, 64.70it/s, loss=0.2046, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 3 (Val MSE: 0.1033)

Época 4/200 [Train] L1=4.71e-04: 100%| | 47/47 [00:00<00:00, 66.55it/s, loss=0.1717, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 4 (Val MSE: 0.0783)

Época 5/200 [Train] L1=4.61e-04: 100%| | 47/47 [00:00<00:00, 65.77it/s, loss=0.1533, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 5 (Val MSE: 0.0672)

Época 5/200 | L1 coef: 4.61e-04

Train - Total: 0.1528 | MSE: 0.0615 | L0: 750.4

Val - Total: 0.1563 | MSE: 0.0672 | L0: 768.0

Best Val MSE: 0.0672 @ época 5 | Sin mejora: 0/15

Época 6/200 [Train] L1=4.52e-04: 100%| | 47/47 [00:00<00:00, 65.92it/s, loss=0.1359, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 6 (Val MSE: 0.0631)

Época 7/200 [Train] L1=4.43e-04: 100%| | 47/47 [00:00<00:00, 63.44it/s, loss=0.1230, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 7 (Val MSE: 0.0596)

Época 8/200 [Train] L1=4.34e-04: 100%| | 47/47 [00:00<00:00, 65.62it/s, loss=0.1152, mse

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 8 (Val MSE: 0.0573)

Época 9/200 [Train] L1=4.25e-04: 100%| | 47/47 [00:00<00:00, 66.52it/s, loss=0.1112, mse

Época 10/200 [Train] L1=4.17e-04: 100%| | 47/47 [00:00<00:00, 64.43it/s, loss=0.1075, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 10 (Val MSE: 0.0552)

Época 10/200 | L1 coef: 4.17e-04

Train - Total: 0.1077 | MSE: 0.0417 | L0: 719.6

Val - Total: 0.1205 | MSE: 0.0552 | L0: 750.3

Best Val MSE: 0.0552 @ época 10 | Sin mejora: 0/15

Época 11/200 [Train] L1=4.09e-04: 100%| | 47/47 [00:00<00:00, 65.05it/s, loss=0.0998, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 11 (Val MSE: 0.0538)

Época 12/200 [Train] L1=4.00e-04: 100%| | 47/47 [00:00<00:00, 66.40it/s, loss=0.0917, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 12 (Val MSE: 0.0522)

Época 13/200 [Train] L1=3.92e-04: 100%| | 47/47 [00:00<00:00, 66.23it/s, loss=0.0944, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 13 (Val MSE: 0.0517)

Época 14/200 [Train] L1=3.85e-04: 100%| | 47/47 [00:00<00:00, 66.47it/s, loss=0.0908, ms
Época 15/200 [Train] L1=3.77e-04: 100%| | 47/47 [00:00<00:00, 65.03it/s, loss=0.0889, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 15 (Val MSE: 0.0511)

Época 15/200 | L1 coef: 3.77e-04
Train - Total: 0.0862 | MSE: 0.0334 | L0: 732.9
Val - Total: 0.1037 | MSE: 0.0511 | L0: 767.3
Best Val MSE: 0.0511 @ época 15 | Sin mejora: 0/15

Época 16/200 [Train] L1=3.69e-04: 100%| | 47/47 [00:00<00:00, 65.21it/s, loss=0.0805, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 16 (Val MSE: 0.0497)

Época 17/200 [Train] L1=3.62e-04: 100%| | 47/47 [00:00<00:00, 66.42it/s, loss=0.0802, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 17 (Val MSE: 0.0495)

Época 18/200 [Train] L1=3.55e-04: 100%| | 47/47 [00:00<00:00, 66.35it/s, loss=0.0786, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 18 (Val MSE: 0.0486)

Época 19/200 [Train] L1=3.48e-04: 100%| | 47/47 [00:00<00:00, 65.32it/s, loss=0.0761, ms
Época 20/200 [Train] L1=3.41e-04: 100%| | 47/47 [00:00<00:00, 66.72it/s, loss=0.0720, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 20 (Val MSE: 0.0482)

Época 20/200 | L1 coef: 3.41e-04
Train - Total: 0.0734 | MSE: 0.0293 | L0: 740.5
Val - Total: 0.0923 | MSE: 0.0482 | L0: 775.5
Best Val MSE: 0.0482 @ época 20 | Sin mejora: 0/15

Época 21/200 [Train] L1=3.34e-04: 100%| | 47/47 [00:00<00:00, 66.94it/s, loss=0.0719, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 21 (Val MSE: 0.0476)

Época 22/200 [Train] L1=3.27e-04: 100%| | 47/47 [00:00<00:00, 67.34it/s, loss=0.0676, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 22 (Val MSE: 0.0474)

Época 23/200 [Train] L1=3.21e-04: 100%| | 47/47 [00:00<00:00, 66.94it/s, loss=0.0655, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 23 (Val MSE: 0.0472)

Época 24/200 [Train] L1=3.14e-04: 100%| | 47/47 [00:00<00:00, 67.27it/s, loss=0.0666, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 24 (Val MSE: 0.0462)

Época 25/200 [Train] L1=3.08e-04: 100%| | 47/47 [00:00<00:00, 68.27it/s, loss=0.0623, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 25 (Val MSE: 0.0459)

Época 25/200 | L1 coef: 3.08e-04
Train - Total: 0.0635 | MSE: 0.0258 | L0: 743.5
Val - Total: 0.0838 | MSE: 0.0459 | L0: 779.6
Best Val MSE: 0.0459 @ época 25 | Sin mejora: 0/15

Época 26/200 [Train] L1=3.02e-04: 100%| | 47/47 [00:00<00:00, 64.39it/s, loss=0.0620, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 26 (Val MSE: 0.0457)

Época 27/200 [Train] L1=2.96e-04: 100%| | 47/47 [00:00<00:00, 65.67it/s, loss=0.0626, ms
Época 28/200 [Train] L1=2.90e-04: 100%| | 47/47 [00:00<00:00, 66.56it/s, loss=0.0578, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 28 (Val MSE: 0.0457)

Época 29/200 [Train] L1=2.84e-04: 100%| | 47/47 [00:00<00:00, 65.54it/s, loss=0.0558, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 29 (Val MSE: 0.0449)

Época 30/200 [Train] L1=2.78e-04: 100%| | 47/47 [00:00<00:00, 62.71it/s, loss=0.0548, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 30 (Val MSE: 0.0445)

Época 30/200 | L1 coef: 2.78e-04
Train - Total: 0.0557 | MSE: 0.0229 | L0: 746.5
Val - Total: 0.0775 | MSE: 0.0445 | L0: 782.6
Best Val MSE: 0.0445 @ época 30 | Sin mejora: 0/15

Época 31/200 [Train] L1=2.73e-04: 100%| | 47/47 [00:00<00:00, 57.71it/s, loss=0.0532, ms
Época 32/200 [Train] L1=2.67e-04: 100%| | 47/47 [00:00<00:00, 57.36it/s, loss=0.0539, ms
Época 33/200 [Train] L1=2.62e-04: 100%| | 47/47 [00:00<00:00, 58.85it/s, loss=0.0536, ms
Época 34/200 [Train] L1=2.57e-04: 100%| | 47/47 [00:00<00:00, 57.90it/s, loss=0.0514, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 34 (Val MSE: 0.0442)

Época 35/200 [Train] L1=2.52e-04: 100%| | 47/47 [00:00<00:00, 58.10it/s, loss=0.0485, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 35 (Val MSE: 0.0436)

Época 35/200 | L1 coef: 2.52e-04
Train - Total: 0.0501 | MSE: 0.0213 | L0: 743.7
Val - Total: 0.0727 | MSE: 0.0436 | L0: 781.8
Best Val MSE: 0.0436 @ época 35 | Sin mejora: 0/15

Época 36/200 [Train] L1=2.47e-04: 100%| | 47/47 [00:00<00:00, 57.49it/s, loss=0.0480, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 36 (Val MSE: 0.0435)

Época 37/200 [Train] L1=2.42e-04: 100%| | 47/47 [00:00<00:00, 54.14it/s, loss=0.0481, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 37 (Val MSE: 0.0434)

Época 38/200 [Train] L1=2.37e-04: 100%| | 47/47 [00:00<00:00, 60.70it/s, loss=0.0456, ms
Época 39/200 [Train] L1=2.32e-04: 100%| | 47/47 [00:00<00:00, 57.70it/s, loss=0.0460, ms
Época 40/200 [Train] L1=2.27e-04: 100%| | 47/47 [00:00<00:00, 57.66it/s, loss=0.0442, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 40 (Val MSE: 0.0432)

Época 40/200 | L1 coef: 2.27e-04
Train - Total: 0.0460 | MSE: 0.0205 | L0: 742.4
Val - Total: 0.0689 | MSE: 0.0432 | L0: 778.4
Best Val MSE: 0.0432 @ época 40 | Sin mejora: 0/15

Época 41/200 [Train] L1=2.23e-04: 100%| | 47/47 [00:00<00:00, 59.68it/s, loss=0.0437, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 41 (Val MSE: 0.0431)

Época 42/200 [Train] L1=2.18e-04: 100%| | 47/47 [00:00<00:00, 60.15it/s, loss=0.0439, ms
Época 43/200 [Train] L1=2.14e-04: 100%| | 47/47 [00:00<00:00, 57.82it/s, loss=0.0431, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 43 (Val MSE: 0.0426)

Época 44/200 [Train] L1=2.10e-04: 100%| | 47/47 [00:00<00:00, 59.47it/s, loss=0.0395, ms
Época 45/200 [Train] L1=2.06e-04: 100%| | 47/47 [00:00<00:00, 57.41it/s, loss=0.0414, ms

Época 45/200 | L1 coef: 2.06e-04
Train - Total: 0.0415 | MSE: 0.0187 | L0: 740.2
Val - Total: 0.0662 | MSE: 0.0433 | L0: 773.9
Best Val MSE: 0.0426 @ época 43 | Sin mejora: 2/15

Época 46/200 [Train] L1=2.01e-04: 100%| | 47/47 [00:00<00:00, 57.47it/s, loss=0.0394, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 46 (Val MSE: 0.0423)

Época 47/200 [Train] L1=1.97e-04: 100%| | 47/47 [00:00<00:00, 59.57it/s, loss=0.0384, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 47 (Val MSE: 0.0416)

Época 48/200 [Train] L1=1.93e-04: 100%| | 47/47 [00:00<00:00, 57.32it/s, loss=0.0372, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 48 (Val MSE: 0.0414)

Época 49/200 [Train] L1=1.90e-04: 100%| | 47/47 [00:00<00:00, 58.40it/s, loss=0.0373, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 49 (Val MSE: 0.0413)

Época 50/200 [Train] L1=1.86e-04: 100%| | 47/47 [00:00<00:00, 60.43it/s, loss=0.0352, ms

Época 50/200 | L1 coef: 1.86e-04
Train - Total: 0.0365 | MSE: 0.0161 | L0: 740.1
Val - Total: 0.0618 | MSE: 0.0414 | L0: 773.7
Best Val MSE: 0.0413 @ época 49 | Sin mejora: 1/15

Época 51/200 [Train] L1=1.82e-04: 100%| | 47/47 [00:00<00:00, 61.26it/s, loss=0.0347, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 51 (Val MSE: 0.0412)

Época 52/200 [Train] L1=1.78e-04: 100%| | 47/47 [00:00<00:00, 59.27it/s, loss=0.0343, ms
Época 53/200 [Train] L1=1.75e-04: 100%| | 47/47 [00:00<00:00, 59.91it/s, loss=0.0344, ms
Época 54/200 [Train] L1=1.71e-04: 100%| | 47/47 [00:00<00:00, 66.09it/s, loss=0.0347, ms
Época 55/200 [Train] L1=1.68e-04: 100%| | 47/47 [00:00<00:00, 63.60it/s, loss=0.0329, ms

Época 55/200 | L1 coef: 1.68e-04
Train - Total: 0.0346 | MSE: 0.0163 | L0: 734.3
Val - Total: 0.0599 | MSE: 0.0415 | L0: 767.0
Best Val MSE: 0.0412 @ época 51 | Sin mejora: 4/15

Época 56/200 [Train] L1=1.65e-04: 100%| | 47/47 [00:00<00:00, 66.07it/s, loss=0.0333, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 56 (Val MSE: 0.0408)

Época 57/200 [Train] L1=1.61e-04: 100%| | 47/47 [00:00<00:00, 66.54it/s, loss=0.0316, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 57 (Val MSE: 0.0406)

Época 58/200 [Train] L1=1.58e-04: 100%| | 47/47 [00:00<00:00, 64.25it/s, loss=0.0318, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 58 (Val MSE: 0.0406)

Época 59/200 [Train] L1=1.55e-04: 100%| | 47/47 [00:00<00:00, 65.71it/s, loss=0.0297, ms

Época 60/200 [Train] L1=1.52e-04: 100%| | 47/47 [00:00<00:00, 67.01it/s, loss=0.0309, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 60 (Val MSE: 0.0403)

Época 60/200 | L1 coef: 1.52e-04

Train - Total: 0.0313 | MSE: 0.0147 | L0: 734.6

Val - Total: 0.0569 | MSE: 0.0403 | L0: 765.0

Best Val MSE: 0.0403 @ época 60 | Sin mejora: 0/15

Época 61/200 [Train] L1=1.49e-04: 100%| | 47/47 [00:00<00:00, 66.03it/s, loss=0.0290, ms

Época 62/200 [Train] L1=1.46e-04: 100%| | 47/47 [00:00<00:00, 64.72it/s, loss=0.0303, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 62 (Val MSE: 0.0398)

Época 63/200 [Train] L1=1.43e-04: 100%| | 47/47 [00:00<00:00, 65.06it/s, loss=0.0280, ms

Época 64/200 [Train] L1=1.40e-04: 100%| | 47/47 [00:00<00:00, 66.91it/s, loss=0.0294, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 64 (Val MSE: 0.0395)

Época 65/200 [Train] L1=1.37e-04: 100% | 47/47 [00:00<00:00, 64.27it/s, loss=0.0274, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 65 (Val MSE: 0.0393)

Época 65/200 | L1 coef: 1.37e-04
Train - Total: 0.0286 | MSE: 0.0137 | L0: 731.9
Val - Total: 0.0543 | MSE: 0.0393 | L0: 761.7
Best Val MSE: 0.0393 @ época 65 | Sin mejora: 0/15

Época 66/200 [Train] L1=1.34e-04: 100% | 47/47 [00:00<00:00, 66.73it/s, loss=0.0295, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 66 (Val MSE: 0.0389)

Época 67/200 [Train] L1=1.32e-04: 100% | 47/47 [00:00<00:00, 66.37it/s, loss=0.0274, ms
Época 68/200 [Train] L1=1.29e-04: 100% | 47/47 [00:00<00:00, 66.46it/s, loss=0.0268, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 68 (Val MSE: 0.0388)

Época 69/200 [Train] L1=1.27e-04: 100% | 47/47 [00:00<00:00, 63.01it/s, loss=0.0256, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 69 (Val MSE: 0.0385)

Época 70/200 [Train] L1=1.24e-04: 100% | 47/47 [00:00<00:00, 66.11it/s, loss=0.0252, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 70 (Val MSE: 0.0385)

Época 70/200 | L1 coef: 1.24e-04
Train - Total: 0.0263 | MSE: 0.0126 | L0: 730.7
Val - Total: 0.0521 | MSE: 0.0385 | L0: 757.0
Best Val MSE: 0.0385 @ época 70 | Sin mejora: 0/15

Época 71/200 [Train] L1=1.22e-04: 100%| | 47/47 [00:00<00:00, 66.37it/s, loss=0.0257, ms
Época 72/200 [Train] L1=1.19e-04: 100%| | 47/47 [00:00<00:00, 64.81it/s, loss=0.0251, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 72 (Val MSE: 0.0380)

Época 73/200 [Train] L1=1.17e-04: 100%| | 47/47 [00:00<00:00, 61.86it/s, loss=0.0247, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 73 (Val MSE: 0.0380)

Época 74/200 [Train] L1=1.14e-04: 100%| | 47/47 [00:00<00:00, 55.11it/s, loss=0.0239, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 74 (Val MSE: 0.0376)

Época 75/200 [Train] L1=1.12e-04: 100%| | 47/47 [00:00<00:00, 64.32it/s, loss=0.0247, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 75 (Val MSE: 0.0371)

Época 75/200 | L1 coef: 1.12e-04
Train - Total: 0.0241 | MSE: 0.0117 | L0: 731.3
Val - Total: 0.0495 | MSE: 0.0371 | L0: 758.0
Best Val MSE: 0.0371 @ época 75 | Sin mejora: 0/15

Época 76/200 [Train] L1=1.10e-04: 100%| | 47/47 [00:00<00:00, 64.48it/s, loss=0.0228, ms
Época 77/200 [Train] L1=1.08e-04: 100%| | 47/47 [00:00<00:00, 66.67it/s, loss=0.0242, ms
Época 78/200 [Train] L1=1.06e-04: 100%| | 47/47 [00:00<00:00, 67.57it/s, loss=0.0232, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 78 (Val MSE: 0.0370)

Época 79/200 [Train] L1=1.03e-04: 100%| | 47/47 [00:00<00:00, 65.98it/s, loss=0.0227, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 79 (Val MSE: 0.0367)

Época 80/200 [Train] L1=1.01e-04: 100%| | 47/47 [00:00<00:00, 65.00it/s, loss=0.0215, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 80 (Val MSE: 0.0360)

Época 80/200 | L1 coef: 1.01e-04
Train - Total: 0.0225 | MSE: 0.0111 | L0: 728.9
Val - Total: 0.0473 | MSE: 0.0360 | L0: 753.9
Best Val MSE: 0.0360 @ época 80 | Sin mejora: 0/15

Época 81/200 [Train] L1=9.93e-05: 100%| | 47/47 [00:00<00:00, 66.37it/s, loss=0.0218, ms
Época 82/200 [Train] L1=9.73e-05: 100%| | 47/47 [00:00<00:00, 67.43it/s, loss=0.0223, ms
Época 83/200 [Train] L1=9.54e-05: 100%| | 47/47 [00:00<00:00, 67.31it/s, loss=0.0205, ms
Época 84/200 [Train] L1=9.35e-05: 100%| | 47/47 [00:00<00:00, 64.14it/s, loss=0.0216, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 84 (Val MSE: 0.0358)

Época 85/200 [Train] L1=9.16e-05: 100%| | 47/47 [00:00<00:00, 67.10it/s, loss=0.0207, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 85 (Val MSE: 0.0349)

Época 85/200 | L1 coef: 9.16e-05
Train - Total: 0.0215 | MSE: 0.0111 | L0: 728.5
Val - Total: 0.0453 | MSE: 0.0349 | L0: 753.8
Best Val MSE: 0.0349 @ época 85 | Sin mejora: 0/15

Época 86/200 [Train] L1=8.98e-05: 100%| | 47/47 [00:00<00:00, 66.96it/s, loss=0.0201, ms
Época 87/200 [Train] L1=8.80e-05: 100%| | 47/47 [00:00<00:00, 66.55it/s, loss=0.0192, ms
Época 88/200 [Train] L1=8.62e-05: 100%| | 47/47 [00:00<00:00, 64.24it/s, loss=0.0188, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 88 (Val MSE: 0.0342)

Época 89/200 [Train] L1=8.45e-05: 100%| | 47/47 [00:00<00:00, 66.30it/s, loss=0.0178, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 89 (Val MSE: 0.0338)

Época 90/200 [Train] L1=8.28e-05: 100%| | 47/47 [00:00<00:00, 66.98it/s, loss=0.0192, ms

Época 90/200 | L1 coef: 8.28e-05
Train - Total: 0.0191 | MSE: 0.0096 | L0: 730.6
Val - Total: 0.0434 | MSE: 0.0339 | L0: 750.9
Best Val MSE: 0.0338 @ época 89 | Sin mejora: 1/15

Época 91/200 [Train] L1=8.12e-05: 100%| | 47/47 [00:00<00:00, 66.39it/s, loss=0.0182, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 91 (Val MSE: 0.0332)

Época 92/200 [Train] L1=7.95e-05: 100%| | 47/47 [00:00<00:00, 67.41it/s, loss=0.0178, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 92 (Val MSE: 0.0330)

Época 93/200 [Train] L1=7.79e-05: 100% | 47/47 [00:00<00:00, 66.53it/s, loss=0.0176, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 93 (Val MSE: 0.0329)

Época 94/200 [Train] L1=7.64e-05: 100% | 47/47 [00:00<00:00, 63.55it/s, loss=0.0175, ms
Época 95/200 [Train] L1=7.49e-05: 100% | 47/47 [00:00<00:00, 66.91it/s, loss=0.0168, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 95 (Val MSE: 0.0326)

Época 95/200 | L1 coef: 7.49e-05
Train - Total: 0.0180 | MSE: 0.0092 | L0: 731.1
Val - Total: 0.0414 | MSE: 0.0326 | L0: 749.5
Best Val MSE: 0.0326 @ época 95 | Sin mejora: 0/15

Época 96/200 [Train] L1=7.34e-05: 100% | 47/47 [00:00<00:00, 66.69it/s, loss=0.0172, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 96 (Val MSE: 0.0321)

Época 97/200 [Train] L1=7.19e-05: 100% | 47/47 [00:00<00:00, 66.10it/s, loss=0.0176, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 97 (Val MSE: 0.0320)

Época 98/200 [Train] L1=7.05e-05: 100% | 47/47 [00:00<00:00, 65.87it/s, loss=0.0169, ms

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 98 (Val MSE: 0.0319)

Época 99/200 [Train] L1=6.90e-05: 100%| | 47/47 [00:00<00:00, 66.65it/s, loss=0.0165, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 99 (Val MSE: 0.0316)

Época 100/200 [Train] L1=6.77e-05: 100%| | 47/47 [00:00<00:00, 63.92it/s, loss=0.0166, m

Época 100/200 | L1 coef: 6.77e-05

Train - Total: 0.0168 | MSE: 0.0088 | L0: 731.7

Val - Total: 0.0397 | MSE: 0.0317 | L0: 749.9

Best Val MSE: 0.0316 @ época 99 | Sin mejora: 1/15

Época 101/200 [Train] L1=6.63e-05: 100%| | 47/47 [00:00<00:00, 65.22it/s, loss=0.0157, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 101 (Val MSE: 0.0308)

Época 102/200 [Train] L1=6.50e-05: 100%| | 47/47 [00:00<00:00, 66.85it/s, loss=0.0155, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 102 (Val MSE: 0.0302)

Época 103/200 [Train] L1=6.37e-05: 100%| | 47/47 [00:00<00:00, 67.55it/s, loss=0.0156, m

Época 104/200 [Train] L1=6.24e-05: 100%| | 47/47 [00:00<00:00, 67.10it/s, loss=0.0148, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 104 (Val MSE: 0.0297)

Época 105/200 [Train] L1=6.12e-05: 100%| | 47/47 [00:00<00:00, 65.71it/s, loss=0.0135, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 105 (Val MSE: 0.0294)

Época 105/200 | L1 coef: 6.12e-05
Train - Total: 0.0155 | MSE: 0.0080 | L0: 732.3
Val - Total: 0.0368 | MSE: 0.0294 | L0: 750.2
Best Val MSE: 0.0294 @ época 105 | Sin mejora: 0/15

Época 106/200 [Train] L1=5.99e-05: 100% | 47/47 [00:00<00:00, 64.94it/s, loss=0.0140, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 106 (Val MSE: 0.0290)

Época 107/200 [Train] L1=5.87e-05: 100% | 47/47 [00:00<00:00, 64.48it/s, loss=0.0141, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 107 (Val MSE: 0.0287)

Época 108/200 [Train] L1=5.76e-05: 100% | 47/47 [00:00<00:00, 66.40it/s, loss=0.0148, m
Época 109/200 [Train] L1=5.64e-05: 100% | 47/47 [00:00<00:00, 65.28it/s, loss=0.0137, m
Época 110/200 [Train] L1=5.53e-05: 100% | 47/47 [00:00<00:00, 66.34it/s, loss=0.0140, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 110 (Val MSE: 0.0283)

Época 110/200 | L1 coef: 5.53e-05
Train - Total: 0.0145 | MSE: 0.0077 | L0: 734.6
Val - Total: 0.0351 | MSE: 0.0283 | L0: 748.6
Best Val MSE: 0.0283 @ época 110 | Sin mejora: 0/15

Época 111/200 [Train] L1=5.42e-05: 100% | 47/47 [00:00<00:00, 65.97it/s, loss=0.0142, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 111 (Val MSE: 0.0279)

Época 112/200 [Train] L1=5.31e-05: 100%| | 47/47 [00:00<00:00, 66.81it/s, loss=0.0144, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 112 (Val MSE: 0.0275)

Época 113/200 [Train] L1=5.20e-05: 100%| | 47/47 [00:00<00:00, 65.47it/s, loss=0.0141, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 113 (Val MSE: 0.0272)

Época 114/200 [Train] L1=5.10e-05: 100%| | 47/47 [00:00<00:00, 66.75it/s, loss=0.0137, m

Época 115/200 [Train] L1=5.00e-05: 100%| | 47/47 [00:00<00:00, 65.63it/s, loss=0.0140, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 115 (Val MSE: 0.0266)

Época 115/200 | L1 coef: 5.00e-05

Train - Total: 0.0142 | MSE: 0.0078 | L0: 733.3

Val - Total: 0.0330 | MSE: 0.0266 | L0: 747.2

Best Val MSE: 0.0266 @ época 115 | Sin mejora: 0/15

Época 116/200 [Train] L1=4.90e-05: 100%| | 47/47 [00:00<00:00, 66.17it/s, loss=0.0135, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 116 (Val MSE: 0.0261)

Época 117/200 [Train] L1=4.80e-05: 100%| | 47/47 [00:00<00:00, 62.75it/s, loss=0.0129, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 117 (Val MSE: 0.0260)

Época 118/200 [Train] L1=4.70e-05: 100%| | 47/47 [00:00<00:00, 66.90it/s, loss=0.0128, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 118 (Val MSE: 0.0255)

Época 119/200 [Train] L1=4.61e-05: 100% | 47/47 [00:00<00:00, 67.36it/s, loss=0.0123, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 119 (Val MSE: 0.0250)

Época 120/200 [Train] L1=4.52e-05: 100% | 47/47 [00:00<00:00, 64.63it/s, loss=0.0130, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 120 (Val MSE: 0.0248)

Época 120/200 | L1 coef: 4.52e-05
Train - Total: 0.0130 | MSE: 0.0071 | L0: 735.1
Val - Total: 0.0307 | MSE: 0.0248 | L0: 746.7
Best Val MSE: 0.0248 @ época 120 | Sin mejora: 0/15

Época 121/200 [Train] L1=4.43e-05: 100% | 47/47 [00:00<00:00, 65.71it/s, loss=0.0125, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 121 (Val MSE: 0.0242)

Época 122/200 [Train] L1=4.34e-05: 100% | 47/47 [00:00<00:00, 65.94it/s, loss=0.0113, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 122 (Val MSE: 0.0236)

Época 123/200 [Train] L1=4.25e-05: 100% | 47/47 [00:00<00:00, 65.46it/s, loss=0.0115, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 123 (Val MSE: 0.0233)

Época 124/200 [Train] L1=4.17e-05: 100%| | 47/47 [00:00<00:00, 65.82it/s, loss=0.0109, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 124 (Val MSE: 0.0229)

Época 125/200 [Train] L1=4.08e-05: 100%| | 47/47 [00:00<00:00, 66.55it/s, loss=0.0099, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 125 (Val MSE: 0.0227)

Época 125/200 | L1 coef: 4.08e-05

Train - Total: 0.0114 | MSE: 0.0059 | L0: 737.3

Val - Total: 0.0282 | MSE: 0.0227 | L0: 746.3

Best Val MSE: 0.0227 @ época 125 | Sin mejora: 0/15

Época 126/200 [Train] L1=4.00e-05: 100%| | 47/47 [00:00<00:00, 64.90it/s, loss=0.0104, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 126 (Val MSE: 0.0223)

Época 127/200 [Train] L1=3.92e-05: 100%| | 47/47 [00:00<00:00, 64.37it/s, loss=0.0105, m

Época 128/200 [Train] L1=3.84e-05: 100%| | 47/47 [00:00<00:00, 65.98it/s, loss=0.0099, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 128 (Val MSE: 0.0220)

Época 129/200 [Train] L1=3.77e-05: 100%| | 47/47 [00:00<00:00, 65.52it/s, loss=0.0095, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 129 (Val MSE: 0.0217)

Época 130/200 [Train] L1=3.69e-05: 100%| | 47/47 [00:00<00:00, 66.79it/s, loss=0.0100, m

Época 130/200 | L1 coef: 3.69e-05
Train - Total: 0.0106 | MSE: 0.0055 | L0: 738.2
Val - Total: 0.0268 | MSE: 0.0217 | L0: 744.8
Best Val MSE: 0.0217 @ época 129 | Sin mejora: 1/15

Época 131/200 [Train] L1=3.62e-05: 100%| | 47/47 [00:00<00:00, 66.92it/s, loss=0.0096, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 131 (Val MSE: 0.0211)

Época 132/200 [Train] L1=3.54e-05: 100%| | 47/47 [00:00<00:00, 64.74it/s, loss=0.0095, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 132 (Val MSE: 0.0208)

Época 133/200 [Train] L1=3.47e-05: 100%| | 47/47 [00:00<00:00, 66.23it/s, loss=0.0097, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 133 (Val MSE: 0.0206)

Época 134/200 [Train] L1=3.40e-05: 100%| | 47/47 [00:00<00:00, 67.02it/s, loss=0.0087, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 134 (Val MSE: 0.0203)

Época 135/200 [Train] L1=3.34e-05: 100%| | 47/47 [00:00<00:00, 63.94it/s, loss=0.0115, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 135 (Val MSE: 0.0201)

Época 135/200 | L1 coef: 3.34e-05
Train - Total: 0.0099 | MSE: 0.0052 | L0: 735.1
Val - Total: 0.0249 | MSE: 0.0201 | L0: 744.3
Best Val MSE: 0.0201 @ época 135 | Sin mejora: 0/15

Época 136/200 [Train] L1=3.27e-05: 100%| | 47/47 [00:00<00:00, 66.61it/s, loss=0.0102, m
Época 137/200 [Train] L1=3.20e-05: 100%| | 47/47 [00:00<00:00, 66.39it/s, loss=0.0091, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 137 (Val MSE: 0.0199)

Época 138/200 [Train] L1=3.14e-05: 100%| | 47/47 [00:00<00:00, 67.29it/s, loss=0.0092, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 138 (Val MSE: 0.0194)

Época 139/200 [Train] L1=3.08e-05: 100%| | 47/47 [00:00<00:00, 64.07it/s, loss=0.0098, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 139 (Val MSE: 0.0185)

Época 140/200 [Train] L1=3.02e-05: 100%| | 47/47 [00:00<00:00, 64.57it/s, loss=0.0084, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 140 (Val MSE: 0.0182)

Época 140/200 | L1 coef: 3.02e-05
Train - Total: 0.0096 | MSE: 0.0052 | L0: 731.2
Val - Total: 0.0227 | MSE: 0.0182 | L0: 738.3
Best Val MSE: 0.0182 @ época 140 | Sin mejora: 0/15

Época 141/200 [Train] L1=2.96e-05: 100%| | 47/47 [00:00<00:00, 65.30it/s, loss=0.0097, m
Época 142/200 [Train] L1=2.90e-05: 100%| | 47/47 [00:00<00:00, 64.96it/s, loss=0.0080, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 142 (Val MSE: 0.0177)

Época 143/200 [Train] L1=2.84e-05: 100%| | 47/47 [00:00<00:00, 64.64it/s, loss=0.0077, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 143 (Val MSE: 0.0174)

Época 144/200 [Train] L1=2.78e-05: 100% | 47/47 [00:00<00:00, 64.82it/s, loss=0.0081, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 144 (Val MSE: 0.0169)

Época 145/200 [Train] L1=2.73e-05: 100% | 47/47 [00:00<00:00, 64.92it/s, loss=0.0077, m

Época 145/200 | L1 coef: 2.73e-05
Train - Total: 0.0083 | MSE: 0.0042 | L0: 731.0
Val - Total: 0.0211 | MSE: 0.0170 | L0: 735.8
Best Val MSE: 0.0169 @ época 144 | Sin mejora: 1/15

Época 146/200 [Train] L1=2.67e-05: 100% | 47/47 [00:00<00:00, 66.28it/s, loss=0.0077, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 146 (Val MSE: 0.0162)

Época 147/200 [Train] L1=2.62e-05: 100% | 47/47 [00:00<00:00, 62.26it/s, loss=0.0082, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 147 (Val MSE: 0.0161)

Época 148/200 [Train] L1=2.57e-05: 100% | 47/47 [00:00<00:00, 64.87it/s, loss=0.0071, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 148 (Val MSE: 0.0159)

Época 149/200 [Train] L1=2.51e-05: 100% | 47/47 [00:00<00:00, 66.09it/s, loss=0.0087, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 149 (Val MSE: 0.0157)

Época 150/200 [Train] L1=2.46e-05: 100%| | 47/47 [00:00<00:00, 66.73it/s, loss=0.0075, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 150 (Val MSE: 0.0156)

Época 150/200 | L1 coef: 2.46e-05
Train - Total: 0.0079 | MSE: 0.0041 | L0: 728.8
Val - Total: 0.0194 | MSE: 0.0156 | L0: 732.5
Best Val MSE: 0.0156 @ época 150 | Sin mejora: 0/15

Época 151/200 [Train] L1=2.41e-05: 100%| | 47/47 [00:00<00:00, 64.01it/s, loss=0.0070, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 151 (Val MSE: 0.0151)

Época 152/200 [Train] L1=2.37e-05: 100%| | 47/47 [00:00<00:00, 65.25it/s, loss=0.0067, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 152 (Val MSE: 0.0149)

Época 153/200 [Train] L1=2.32e-05: 100%| | 47/47 [00:00<00:00, 64.90it/s, loss=0.0072, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 153 (Val MSE: 0.0149)

Época 154/200 [Train] L1=2.27e-05: 100%| | 47/47 [00:00<00:00, 61.95it/s, loss=0.0063, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 154 (Val MSE: 0.0144)

Época 155/200 [Train] L1=2.23e-05: 100%| | 47/47 [00:00<00:00, 65.70it/s, loss=0.0066, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 155 (Val MSE: 0.0142)

Época 155/200 | L1 coef: 2.23e-05

Train - Total: 0.0071 | MSE: 0.0036 | L0: 726.1

Val - Total: 0.0177 | MSE: 0.0142 | L0: 731.0

Best Val MSE: 0.0142 @ época 155 | Sin mejora: 0/15

Época 156/200 [Train] L1=2.18e-05: 100%| | 47/47 [00:00<00:00, 65.50it/s, loss=0.0064, m

Época 157/200 [Train] L1=2.14e-05: 100%| | 47/47 [00:00<00:00, 65.34it/s, loss=0.0062, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 157 (Val MSE: 0.0138)

Época 158/200 [Train] L1=2.10e-05: 100%| | 47/47 [00:00<00:00, 64.16it/s, loss=0.0064, m

Época 159/200 [Train] L1=2.05e-05: 100%| | 47/47 [00:00<00:00, 66.41it/s, loss=0.0069, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 159 (Val MSE: 0.0136)

Época 160/200 [Train] L1=2.01e-05: 100%| | 47/47 [00:00<00:00, 64.27it/s, loss=0.0070, m

Época 160/200 | L1 coef: 2.01e-05

Train - Total: 0.0072 | MSE: 0.0040 | L0: 723.4

Val - Total: 0.0170 | MSE: 0.0138 | L0: 727.0

Best Val MSE: 0.0136 @ época 159 | Sin mejora: 1/15

Época 161/200 [Train] L1=1.97e-05: 100%| | 47/47 [00:00<00:00, 57.98it/s, loss=0.0058, m

Época 162/200 [Train] L1=1.93e-05: 100%| | 47/47 [00:00<00:00, 59.52it/s, loss=0.0068, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 162 (Val MSE: 0.0133)

Época 163/200 [Train] L1=1.89e-05: 100%| | 47/47 [00:00<00:00, 58.64it/s, loss=0.0065, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 163 (Val MSE: 0.0130)

Época 164/200 [Train] L1=1.86e-05: 100%| | 47/47 [00:00<00:00, 60.97it/s, loss=0.0060, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 164 (Val MSE: 0.0126)

Época 165/200 [Train] L1=1.82e-05: 100%| | 47/47 [00:00<00:00, 64.28it/s, loss=0.0057, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 165 (Val MSE: 0.0123)

Época 165/200 | L1 coef: 1.82e-05

Train - Total: 0.0063 | MSE: 0.0033 | L0: 721.1

Val - Total: 0.0153 | MSE: 0.0123 | L0: 723.5

Best Val MSE: 0.0123 @ época 165 | Sin mejora: 0/15

Época 166/200 [Train] L1=1.78e-05: 100%| | 47/47 [00:00<00:00, 66.23it/s, loss=0.0063, m

Época 167/200 [Train] L1=1.75e-05: 100%| | 47/47 [00:00<00:00, 65.98it/s, loss=0.0058, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 167 (Val MSE: 0.0120)

Época 168/200 [Train] L1=1.71e-05: 100%| | 47/47 [00:00<00:00, 64.95it/s, loss=0.0050, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 168 (Val MSE: 0.0115)

Época 169/200 [Train] L1=1.68e-05: 100%| | 47/47 [00:00<00:00, 65.51it/s, loss=0.0052, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 169 (Val MSE: 0.0112)

Época 170/200 [Train] L1=1.65e-05: 100%| | 47/47 [00:00<00:00, 64.91it/s, loss=0.0051, m

Época 170/200 | L1 coef: 1.65e-05
Train - Total: 0.0056 | MSE: 0.0028 | L0: 720.0
Val - Total: 0.0140 | MSE: 0.0112 | L0: 721.7
Best Val MSE: 0.0112 @ época 169 | Sin mejora: 1/15

Época 171/200 [Train] L1=1.61e-05: 100%| | 47/47 [00:00<00:00, 66.20it/s, loss=0.0051, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 171 (Val MSE: 0.0110)

Época 172/200 [Train] L1=1.58e-05: 100%| | 47/47 [00:00<00:00, 65.39it/s, loss=0.0051, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 172 (Val MSE: 0.0108)

Época 173/200 [Train] L1=1.55e-05: 100%| | 47/47 [00:00<00:00, 63.74it/s, loss=0.0051, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 173 (Val MSE: 0.0105)

Época 174/200 [Train] L1=1.52e-05: 100%| | 47/47 [00:00<00:00, 66.38it/s, loss=0.0057, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 174 (Val MSE: 0.0105)

Época 175/200 [Train] L1=1.49e-05: 100%| | 47/47 [00:00<00:00, 66.71it/s, loss=0.0050, m

Época 175/200 | L1 coef: 1.49e-05
Train - Total: 0.0055 | MSE: 0.0029 | L0: 719.1
Val - Total: 0.0136 | MSE: 0.0110 | L0: 719.5
Best Val MSE: 0.0105 @ época 174 | Sin mejora: 1/15

Época 176/200 [Train] L1=1.46e-05: 100%| | 47/47 [00:00<00:00, 65.15it/s, loss=0.0052, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 176 (Val MSE: 0.0104)

Época 177/200 [Train] L1=1.43e-05: 100%| | 47/47 [00:00<00:00, 66.58it/s, loss=0.0051, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 177 (Val MSE: 0.0103)

Época 178/200 [Train] L1=1.40e-05: 100%| | 47/47 [00:00<00:00, 65.83it/s, loss=0.0046, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 178 (Val MSE: 0.0101)

Época 179/200 [Train] L1=1.37e-05: 100%| | 47/47 [00:00<00:00, 65.34it/s, loss=0.0046, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 179 (Val MSE: 0.0096)

Época 180/200 [Train] L1=1.34e-05: 100%| | 47/47 [00:00<00:00, 65.48it/s, loss=0.0050, m

Época 180/200 | L1 coef: 1.34e-05
Train - Total: 0.0048 | MSE: 0.0024 | L0: 719.2
Val - Total: 0.0120 | MSE: 0.0097 | L0: 719.2
Best Val MSE: 0.0096 @ época 179 | Sin mejora: 1/15

Época 181/200 [Train] L1=1.32e-05: 100%| | 47/47 [00:00<00:00, 67.16it/s, loss=0.0045, m

Época 182/200 [Train] L1=1.29e-05: 100%| | 47/47 [00:00<00:00, 65.04it/s, loss=0.0041, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 182 (Val MSE: 0.0092)

Época 183/200 [Train] L1=1.27e-05: 100%| | 47/47 [00:00<00:00, 66.54it/s, loss=0.0041, m
Época 184/200 [Train] L1=1.24e-05: 100%| | 47/47 [00:00<00:00, 66.38it/s, loss=0.0046, m
Época 185/200 [Train] L1=1.21e-05: 100%| | 47/47 [00:00<00:00, 64.74it/s, loss=0.0052, m

Época 185/200 | L1 coef: 1.21e-05

Train - Total: 0.0048 | MSE: 0.0026 | L0: 715.4

Val - Total: 0.0116 | MSE: 0.0094 | L0: 717.1

Best Val MSE: 0.0092 @ época 182 | Sin mejora: 3/15

Época 186/200 [Train] L1=1.19e-05: 100%| | 47/47 [00:00<00:00, 65.72it/s, loss=0.0047, m
Época 187/200 [Train] L1=1.17e-05: 100%| | 47/47 [00:00<00:00, 65.97it/s, loss=0.0045, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 187 (Val MSE: 0.0091)

Época 188/200 [Train] L1=1.14e-05: 100%| | 47/47 [00:00<00:00, 66.25it/s, loss=0.0047, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 188 (Val MSE: 0.0088)

Época 189/200 [Train] L1=1.12e-05: 100%| | 47/47 [00:00<00:00, 66.13it/s, loss=0.0044, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 189 (Val MSE: 0.0086)

Época 190/200 [Train] L1=1.10e-05: 100%| | 47/47 [00:00<00:00, 64.90it/s, loss=0.0044, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 190 (Val MSE: 0.0084)

Época 190/200 | L1 coef: 1.10e-05
Train - Total: 0.0045 | MSE: 0.0025 | L0: 716.0
Val - Total: 0.0104 | MSE: 0.0084 | L0: 715.9
Best Val MSE: 0.0084 @ época 190 | Sin mejora: 0/15

Época 191/200 [Train] L1=1.08e-05: 100%| | 47/47 [00:00<00:00, 65.50it/s, loss=0.0039, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 191 (Val MSE: 0.0084)

Época 192/200 [Train] L1=1.05e-05: 100%| | 47/47 [00:00<00:00, 62.69it/s, loss=0.0040, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 192 (Val MSE: 0.0078)

Época 193/200 [Train] L1=1.03e-05: 100%| | 47/47 [00:00<00:00, 66.53it/s, loss=0.0038, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 193 (Val MSE: 0.0077)

Época 194/200 [Train] L1=1.01e-05: 100%| | 47/47 [00:00<00:00, 65.96it/s, loss=0.0038, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt
Mejor modelo guardado en época 194 (Val MSE: 0.0075)

Época 195/200 [Train] L1=9.93e-06: 100%| | 47/47 [00:00<00:00, 65.01it/s, loss=0.0034, m

Época 195/200 | L1 coef: 9.93e-06
Train - Total: 0.0039 | MSE: 0.0021 | L0: 716.5
Val - Total: 0.0094 | MSE: 0.0075 | L0: 717.3
Best Val MSE: 0.0075 @ época 194 | Sin mejora: 1/15

Época 196/200 [Train] L1=9.73e-06: 100%| | 47/47 [00:00<00:00, 67.36it/s, loss=0.0031, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 196 (Val MSE: 0.0074)

Época 197/200 [Train] L1=9.53e-06: 100%| | 47/47 [00:00<00:00, 64.70it/s, loss=0.0033, m

Época 198/200 [Train] L1=9.34e-06: 100%| | 47/47 [00:00<00:00, 63.49it/s, loss=0.0034, m

Época 199/200 [Train] L1=9.16e-06: 100%| | 47/47 [00:00<00:00, 66.69it/s, loss=0.0032, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\500games\sae_standard_l1-decay_l6_500g_ex4_best.pt

Mejor modelo guardado en época 199 (Val MSE: 0.0070)

Época 200/200 [Train] L1=8.97e-06: 100%| | 47/47 [00:00<00:00, 66.43it/s, loss=0.0032, m

Época 200/200 | L1 coef: 8.97e-06

Train - Total: 0.0037 | MSE: 0.0020 | L0: 716.3

Val - Total: 0.0088 | MSE: 0.0071 | L0: 716.2

Best Val MSE: 0.0070 @ época 199 | Sin mejora: 1/15

RESUMEN DEL ENTRENAMIENTO

Mejor modelo en época 199 con Val MSE = 0.0070

Early stopping tras 200 épocas

Métricas Finales (Última Época):

Métrica	Train	Validation
Total Loss	0.0037	0.0088
MSE Loss	0.0020	0.0071
L0 Sparsity	716.3	716.2
Fraction Active	4.37%	4.37%

8. Guardar Modelo

```

# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final'
)
checkpoint_path = CONFIG['save_dir'] / final_model_name
torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

```

```

})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)

```

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\500games\sae_standard_l1-decay_l6_500g_ex4_final.pt

```

=====
CHECKPOINTS GUARDADOS
=====

```

Tipo	Ubicación
Local saved_model\sae_standard_l1-decay_l6_500g_final.pt	
Organizado	layer_06/models/...

```

=====

```

9. Métricas básicas del modelo

```

# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
}

```

```

    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "sparsity_decay": CONFIG["sparsity_decay"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

# Resumen
metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",
        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})

print("" + "="*50)
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)

```

```

=====
METRICAS DE ENTRENAMIENTO GUARDADAS
=====

```

```

      Metrica      Valor
Val MSE (final) 0.007071
Val MSE (mejor) 0.006969
  Val L0 (final)    716.2
Fraccion Activa    4.37%
Archivo: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard
decay_standard\500games\ex4\metrics\training_metrics.json

```

10. Visualización de Resultados

```
=====

# Gráficas de pérdidas (Train vs Validation) + L1 Decay
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8)
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0.8)
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
```

```

axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# Sparsity Coefficient (L1 Decay)
axes[1, 1].plot(history['sparsity_coef_history'], color='brown', linewidth=2)
axes[1, 1].set_title('L1 Coefficient (Decay)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('Sparsity Coef')
axes[1, 1].set_yscale('log')
axes[1, 1].grid(True, alpha=0.3)

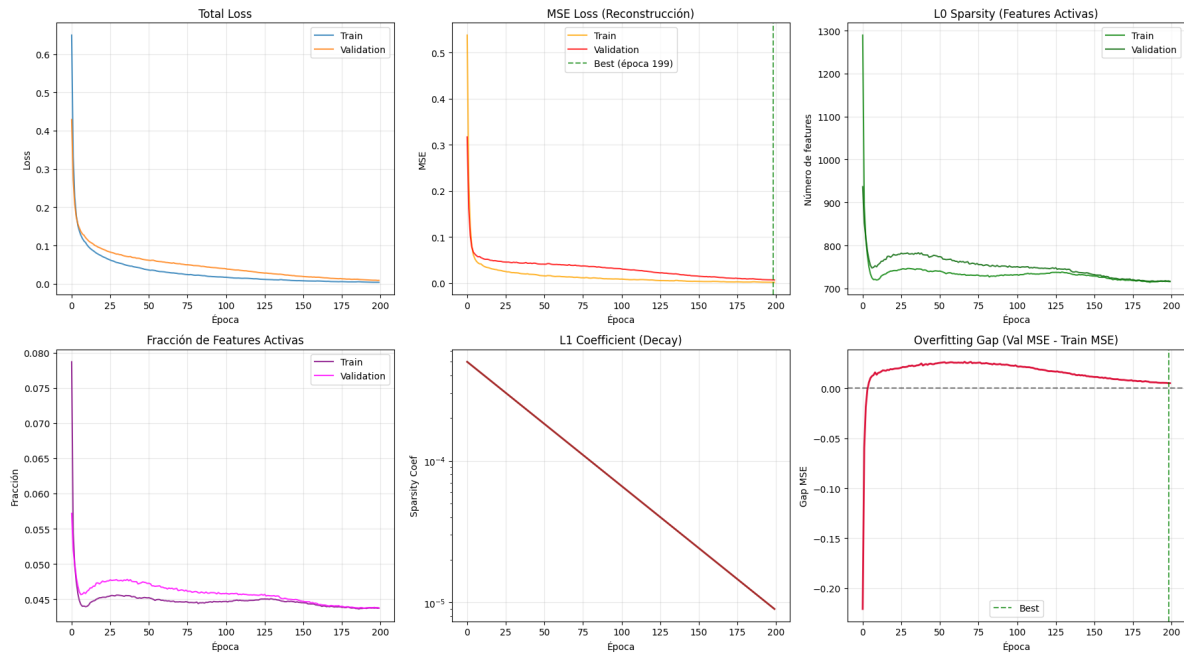
# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('  Gráficas guardadas exitosamente')

print('='*50)

```



=====
 Gráficas guardadas exitosamente
 =====

11. Resumen Final

```
# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:,}"
    ]
})
```

```

print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')
print(final_summary.to_string(index=False))

# Archivos
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)

```

```
print('  ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)
```

```
=====
RESUMEN FINAL DEL ENTRENAMIENTO
=====
```

Arquitectura:

Componente	Valor
Input	512
Hidden	16384 (32.0x)
Parámetros	16,794,112

Datos:

Conjunto	Muestras
Train	23,600
Validation	5,900
Total	29,500

Métricas Finales:

Métrica	Valor
Train Loss	0.0037
Val Loss	0.0088
Train MSE	0.0020
Val MSE	0.0071
Train L0	716.3
Val L0	716.2

Archivos Guardados:

Tipo	Modelo
decay_l6_500g_final.pt	
Gráficas	c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\decay_standard\500games\ex4\plots\training_curves.png

```
=====
ENTRENAMIENTO COMPLETADO EXITOSAMENTE
=====
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_l1-decay.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
```

```
else:  
    print(f' Error al generar PDF:')  
    print(result.stderr)
```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\500games\ex4\sae_standard_l1-decay_l6_500g_ex4_training.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\500games\ex4\sae_standard_l1-decay_l6_500g_ex4_training.pdf